Q.No.	Question Description	Marks
1.	A smart-campus sensor records the number of events detected during successive time intervals. A smart-campus sensor records the number of events detected during successive time intervals. For a simulation experiment, the event count follows a Fibonacci-like sequence in which each new term obtained by adding the previous two terms. The first two terms of the sequence are 2 and 3. Design a Python program to generate the first N terms of this sequence, where N is a positive integer entered by the user. **(a)** Write a Python program using an iterative structure to generate and display the first N terms. Do not use recursion or any built-in function that directly generates the sequence. **(b)** Extend the program to also calculate and display the sum of all the generated terms. **(c)** Test the program for N = 1, N = 2, and N = 7, and show the corresponding output	**5+3+2**
2.	A software system needs to calculate the number of possible arrangements of N distinct objects. The number of arrangements is represented by the factorial of N, denoted by N!, where $$N! = N \times (N-1) \times (N-2) \times \ldots \times 1$$ (Assume:0! = 1). Design a Python program to calculate N!, where N is an integer entered by the user. (a) Design an iterative algorithm using a loop to calculate N!. Do not use Python's `math.factorial()` function or recursion. (b) Modify the program to validate the input before performing the calculation. If the user enters a negative integer, the program must not attempt to calculate its factorial and must display an appropriate error message. For `N = 0`, the program must correctly produce `0!=1`.	**6+4**

3.	A hospital maintains the following patient-monitoring data. Each tuple contains:
	
	`(Patient ID, Heart Rate, Oxygen Level, Body Temperature)`
	
	`patients = [`
	`("P101", 78, 98, 36.8),`
	`("P102", 105, 94, 37.5),`
	`("P103", 72, 97, 36.6),`
	`("P104", 115, 91, 38.2),`
	`("P105", 88, 96, 37.0)`
		3+3+4

	} Write a Python program to analyze the patient data using iteration and list operations. (a) Identify and display all patients whose heart rate is greater than 100 OR oxygen level is below 95. A patient must displayed if either condition or both conditions are satisfied. (b) Create a separate list containing only the patient IDs whose body temperature is strictly greater than 37.5°C. A temperature of exactly 37.5°C must not be included. (c) Find and display the patient(s) having the lowest oxygen level without using `min()`, `sorted()`, or any other built-in function that directly determines the minimum value. Your solution must also correctly handle the case where more than one patient has the same lowest oxygen level.	
4.	Consider the following unsorted list: `numbers = [27, 12, 21, 15, 27, 18, 12, 30, 21, 24, 15]` Design and write a Python program to analyze the list subject to the following requirements: (a) Create a new list containing only the distinct elements, while preserving the order of their first occurrence. Do not use `set()`. (b) Using the distinct elements obtained in (a), determine and display the 4th smallest distinct element without using `sort()`, `sorted()`, or `min()`. (c) If the list contains fewer than four distinct elements, display an appropriate message instead of producing a result. Your program should work for the given list without hard-coding the answer	**5+3+2**

5.	An online shopping system stores product information in a Python dictionary. The product name is the key and the product price is the value.
	
	Consider the following dictionary:
	
	`products = {`
	`"Laptop": 55000,`
	`"Mobile": 25000,`
	`"Headphones": 3500,`
	`"Tablet": 18000,`
	`"Smartwatch": 7000`
	`}`
	Write a Python program to perform the following operations:
	
	(a) Traverse the dictionary to calculate and display the total price of all products. Do not use `sum()`.
	
	(b) Traverse the dictionary and identify and display the product name and price of the most expensive product. Do not use `max()`.
	
	(c) Extend your program to count and display the number of products whose price is greater than ₹20,000.
	
	Your program should determine all results by processing the dictionary;
		4+4+2

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